

The Problem

Company

The Houston Parks & Recreation Department's Urban Forestry Division

Context

The Division Manager of Urban Forestry wants to understand which work zones and contractors are driving overall SLA compliance below target, and is the bottleneck caused by backlog age in specific service types or caused by low workforce capacity?

Problem Statement

Perform root cause analysis of tree maintenance work orders from 2025 to answer the following business questions:

- Why is overall SLA compliance dropping below target?
- Why are tree trimming jobs missing their SLA at a high rate?
- Which work zone is most underperforming on tree trimming? Why?

The Process

Step 1

Create a project design document

- Identify what data sources are available to answer the business questions
- Review the integrity, quality, and limitations of the data

Tools used: Word, Excel

Step 2

Clean raw data from 2020 to 2025

- Clean work order records from the Houston Parks & Recreation Department
- Create dimension and fact tables for star schema in PostgreSQL database

Tools used: Python, SQL, DBeaver

Step 3

Create visualizations for insights

- Create visualizations that answer the business questions
- Use visualizations to construct a dashboard

Tools used: Tableau Public

The Results

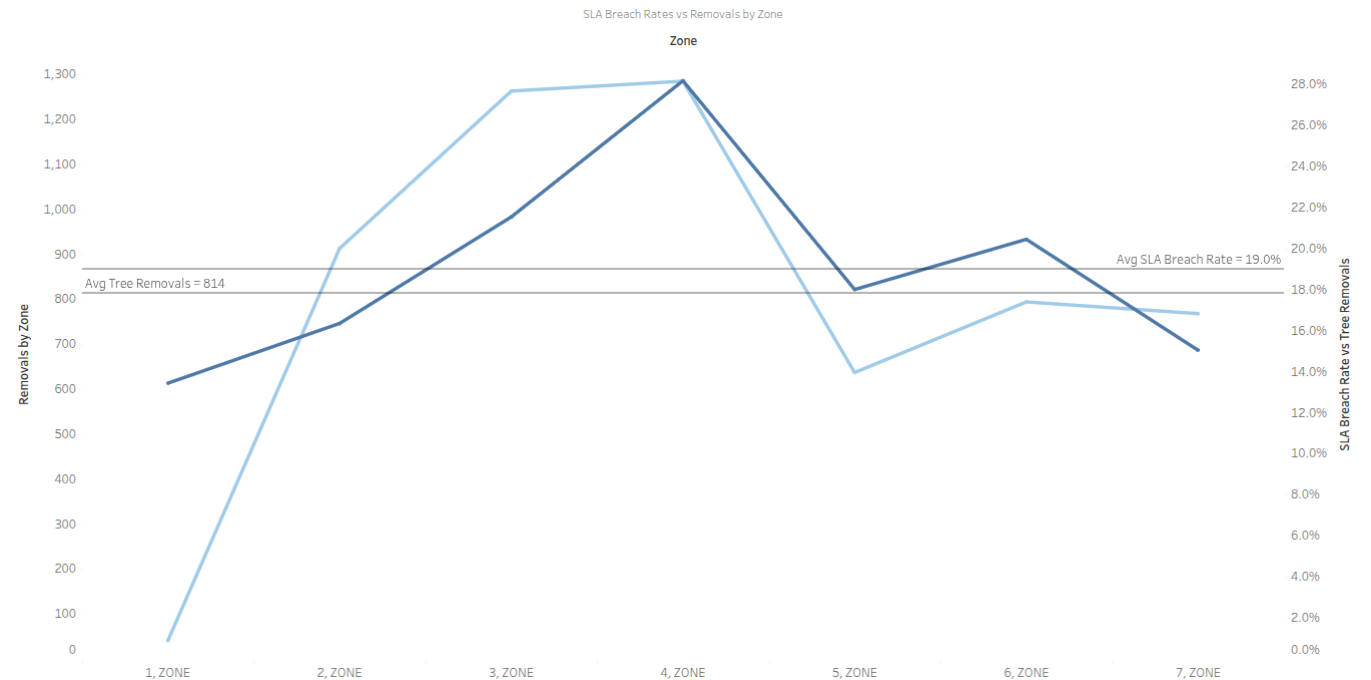
SLA breach rate versus tree removal volume shows positive relationship

Insight:

When reviewing tree removal work orders only, the macro trend shows a clear positive relationship between high removal work orders and increased SLA breach rates.

From Zone 1 through Zone 5, both metrics rise and fall together. When removal volume increases, the SLA breach rate increases proportionally.

There is a slight divergence in Zones 6 & 7 where removal volume drops further while the breach rate flattens out, but the overall positive relationship remains.



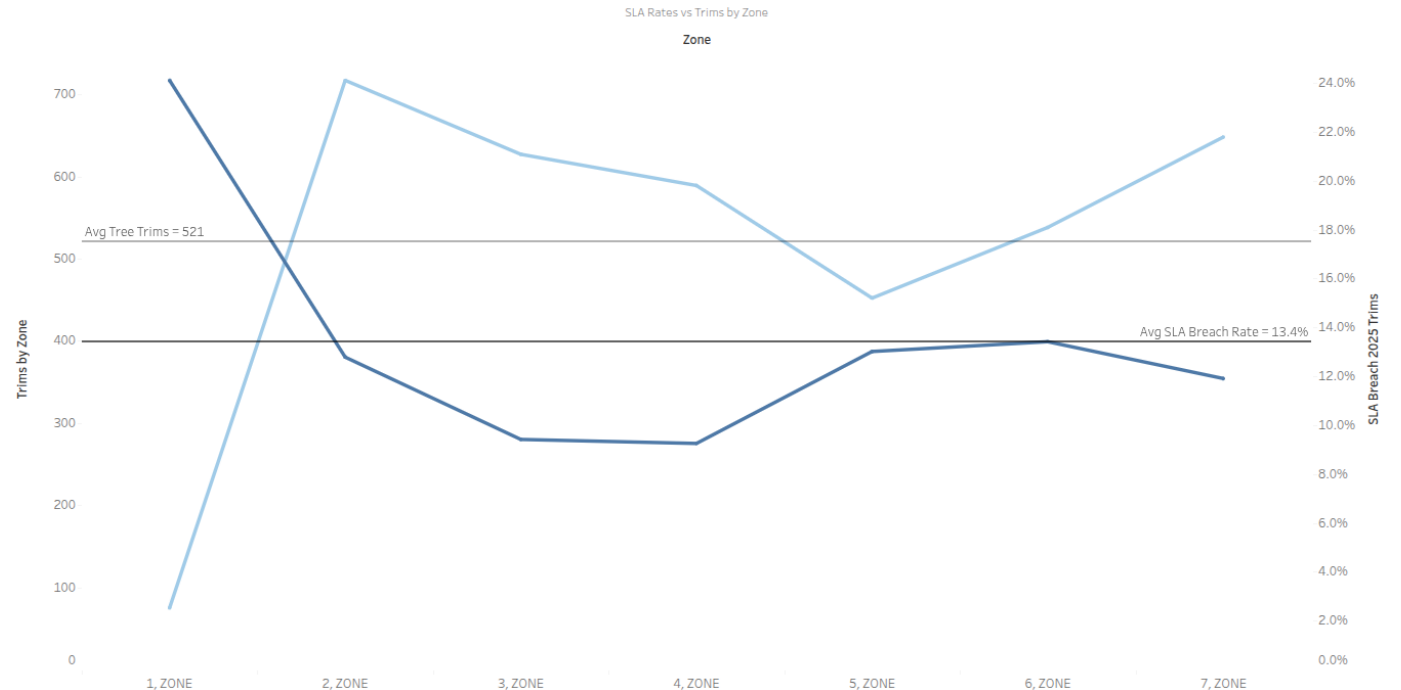
The Results

SLA breach rate versus tree trimming volume shows negative relationship

Insight:

When reviewing tree trimming work orders only, the macro trend shows a clear negative relationship between high trimming work orders and increased SLA breach rates.

Unlike the removal metrics where removal volume and SLA breach rates moved together, higher trimming volumes do not drive up SLA breach rates, which indicates either an underlying operational issue within the department or limitations with the contractors, most notably within Zone 2 where trim volume drops sharply while the SLA breach rate simultaneously spikes to its peak.



The Results

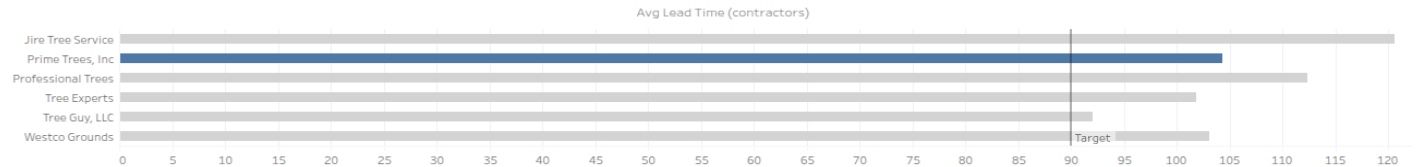
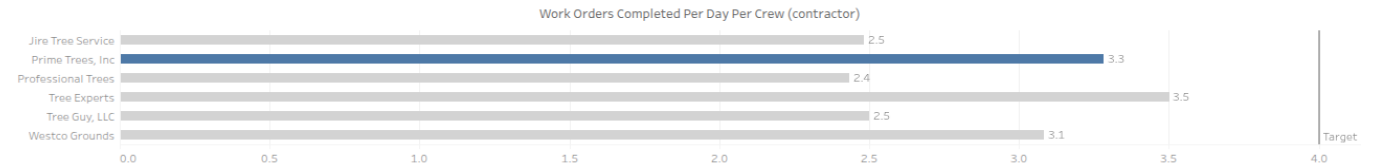
High trim volume overwhelms crew capacity, resulting in long lead times

Insight:

The contractor servicing Zone 2 (Prime Trees, Inc.) places well above most other contractors for work orders completed per day per crew. However, despite having high crew output, Zone 2 experienced a large spike in SLA breach rate while handling a high volume of tree trimming work orders.

This indicates that while Prime Trees is maintaining high daily output per crew, the intake volume of work orders is overwhelming their crew's capacity.

High volume also impacts their overall lead time from intake to completion, where Prime Trees has the 3rd longest lead time amongst all contractors.



The Solution

Adjust crew capacity requirements or reallocate zones to meet demand

Insight:

This root cause analysis proves the bottleneck is a result of high intake volume and highlights the need for resource realignment – either by adjusting crew capacity requirements within the contract or reallocating zones to meet the trimming demand.

The Impact

10.6% reduction in SLA breach rates for tree trimming in Zone 2

Insight:

Using verified daily outputs of 3.3 work orders completed per day per crew, an annual trimming volume of 700 work orders for Zone 2, a 104-day average lead time, and a baseline of 2 active trimming crews in Zone 2, requiring just one additional trimming crew will boost daily output by 50%, from 6.6 work orders completed to 9.9, and lower average lead time by 33.65%, from 104 days to about 69 days, meeting the department's target timeline of 90-days.

These changes will yield a 10.6% reduction in SLA breach rates for tree trimming where most needed in Zone 2.